Markov Chains: 10 Hard Problems

Multi-Step & Conditioning · Classification & Absorption · Periodicity · Stationary & Quant

Real Stage-1 Markov questions rarely say "find P^n" — they hide a closed form, a conditioning trick, a number-theory fact about cycle lengths, or a stationary-vs-return-time identity inside a short story. Each problem below fuses at least two ideas from Chapter 4.1. Every option set has one correct answer and three answers built from a specific, common error: learn to recognize your own mistake in the wrong options.

I. Multi-Step Transitions & Conditioning

Problem H1 — Five steps in a two-state chain, without matrix powers

A two-state chain has $P = \begin{pmatrix} 0.7 & 0.3 \\ 0.2 & 0.8 \end{pmatrix}$ and starts in state 1. Find $P(X_5 = 1)$.

Options: (A) 0.4000 (B) 0.4188 (C) 0.1681 (D) 0.5813

Solution. Do not multiply five matrices. For a two-state chain with $p = P_{12} = 0.3$, $q = P_{21} = 0.2$:

$$\pi = \left(\tfrac{q}{p+q}, \tfrac{p}{p+q}\right) = (0.4,\, 0.6), \qquad \lambda_2 = 1 - p - q = 0.5.$$

$$P^n_{11} = \pi_1 + \pi_2 \lambda_2^{\,n} \;\Rightarrow\; P^5_{11} = 0.4 + 0.6\left(\tfrac{1}{2}\right)^5 = \boxed{0.41875}.$$

Trap analysis: (A) is the stationary value, i.e. $n = \infty$; after only 5 steps the chain still remembers its start (0.019 above π_1). (C) is 0.7^5: it counts only the path that stays in state 1 every step and ignores returning from state 2. (D) is $P(X_5 = 2) = 1 - 0.41875$, the wrong state.

Correct: (B) 0.4188

★ *Intuition:* The chain forgets its starting point at rate $|\lambda_2|^n$, like a hot cup cooling toward room temperature: the gap to π is cut in half (here) every step. After 5 steps the cup is within 0.019 of room temperature but not yet at it.

▶ *Speed insight:* Two states: $P^n_{ii} = \pi_i + (1 - \pi_i)\lambda_2^n$. Memorize $\lambda_2 = 1 - p - q$; it replaces any matrix power.

Problem H2 — Conditioning on the past *and* the future

A chain on $\{A, B, C\}$ has

$$P = \begin{pmatrix} 0.5 & 0.3 & 0.2 \\ 0.2 & 0.6 & 0.2 \\ 0.4 & 0.1 & 0.5 \end{pmatrix}.$$

Given $X_0 = A$ and $X_2 = B$, find $P(X_1 = B \mid X_0 = A, X_2 = B)$.

Options: (A) 0.180 (B) 0.300 (C) 0.514 (D) 0.600

Solution. By Bayes with the Markov property, the middle state j has weight $P_{Aj}P_{jB}$:

$$P(X_1 = j \mid X_0{=}A, X_2{=}B) = \frac{P_{Aj}P_{jB}}{P^2_{AB}},$$

$$P^2_{AB} = 0.5(0.3) + 0.3(0.6) + 0.2(0.1) = 0.35.$$

$$P(X_1 = B \mid \cdots) = \frac{0.3 \times 0.6}{0.35} = \frac{18}{35} \approx \boxed{0.514}.$$

(The three conditional probabilities are 0.429, 0.514, 0.057 for A, B, C and sum to 1.)

Trap analysis: (B) is $P_{AB} = 0.3$: a forward-only answer that ignores that you *know* $X_2 = B$. (D) is $P_{BB} = 0.6$: it ignores that you know $X_0 = A$. (A) is the unnormalized numerator 0.3×0.6 — you must divide by the total probability $P^2_{AB} = 0.35$ of the evidence.

Correct: (C) 0.514

★ *Intuition:* A detective knows the suspect was at the station in the morning (A) and at the bank in the evening (B). Where was he at noon? Each possible stop is scored by the product of the *leg into it* and the *leg out of it*, then rescaled to add to 1 — a "Markov bridge."

▶ *Speed insight:* Conditioning on $X_0 = a, X_2 = b$: weight$(j) = P_{aj}P_{jb}$, normalize by P^2_{ab}. Never just use a one-step row.

Problem H3 — A lazy symmetric chain

Three states; each state stays put with probability 0.4 and moves to each of the other two with probability 0.3. Find $P(X_3 = 1 \mid X_0 = 1)$.

Options: (A) 0.0640 (B) 0.3333 (C) 0.3340 (D) 0.3400

Solution. $P = 0.1\, I + 0.3\, J$ where J is the all-ones matrix, so its eigenvalues are 1 (eigenvector $\mathbf{1}$) and $0.4 - 0.3 = 0.1$ (twice). Hence

$$P^n_{11} = \frac{1}{3} + \frac{2}{3}(0.1)^n \;\Rightarrow\; P^3_{11} = \frac{1}{3} + \frac{2}{3}(0.001) = \boxed{0.3340}.$$

Check at $n = 2$: $0.4^2 + 0.3^2 + 0.3^2 = 0.34 = \frac{1}{3} + \frac{2}{3}(0.01)$. ✓

Trap analysis: (A) is 0.4^3, only the path that stays every step. (B) is the limit $\pi_1 = \frac{1}{3}$, reached only as $n \to \infty$. (D) is $P^2_{11} = 0.34$: the exponent is off by one.

Correct: (C) 0.3340

★ *Intuition:* With a "lazy" symmetric chain the only memory is the extra 0.1 of self-preference, and that memory shrinks tenfold every step — by step 3 the chain is within 0.0007 of perfectly uniform.

▶ *Speed insight:* For $P = aI + bJ$ the nontrivial eigenvalue is a (here 0.1), so convergence is geometric at rate a; $P^n_{ii} = \frac{1}{m} + (1 - \frac{1}{m})a^n$ for m states.

II. Classification, Absorption & Recurrence

Problem H4 — Which closed class eats the chain?

Five states with

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0.2 & 0 & 0.3 & 0.5 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

Starting from state 3, find the probability of eventually landing in the class $\{1, 2\}$.

Options: (A) 0.20 (B) 0.2857 (C) 0.40 (D) 0.50

Solution. State 3 is **transient**: it leaves itself with probability 0.7 per step, into $\{1, 2\}$ w.p. 0.2 or $\{4, 5\}$ w.p. 0.5. The closed classes are $\{1, 2\}$ (aperiodic) and $\{4, 5\}$ (period 2). With $h = P(\text{end in } \{1, 2\})$:

$$h = 0.2 + 0.3\,h \;\Rightarrow\; h = \frac{0.2}{0.7} = \boxed{\tfrac{2}{7} \approx 0.2857}.$$

Trap analysis: (A) is the one-step probability 0.2, ignoring that the chain may loiter at 3 and try again. (C) is $0.2/0.5$ — a ratio of the two exit weights instead of a share of the total exit mass. (D) assumes the two exits are equally likely.

Correct: (B) 0.2857

★ *Intuition:* State 3 is a lobby with two doors and a revolving "stay" option. Each minute you either stay (30%), take door 1 (20%) or door 2 (50%). Staying changes nothing, so the answer is just door 1's share of the *exits*: $\frac{0.2}{0.2+0.5}$.

▶ *Speed insight:* With a self-loop s at a transient state, condition on leaving: exit probabilities are $\frac{P_{3j}}{1-s}$. Matrix powers are a trap here: P^n does not even converge (class $\{4, 5\}$ alternates), yet the class-absorption probability is perfectly well defined.

Problem H5 — Expected time before absorption

A chain on $\{1, 2, 3, 4\}$ has absorbing states 1 and 4 and

$$P_{2\cdot} = (0.3,\ 0.4,\ 0.3,\ 0), \qquad P_{3\cdot} = (0,\ 0.2,\ 0.5,\ 0.3).$$

Starting from state 2, find the expected number of steps until absorption.

Options: (A) 1.67 (B) 2.08 (C) 2.50 (D) 3.33

Solution. Let k_i be the expected time to absorption (first-step analysis):

$$k_2 = 1 + 0.4k_2 + 0.3k_3, \qquad k_3 = 1 + 0.2k_2 + 0.5k_3.$$

From the second, $k_3 = 2 + 0.4k_2$. Substituting, $0.48k_2 = 1.6$:

$$k_2 = \frac{1.6}{0.48} = \boxed{\tfrac{10}{3} \approx 3.33}.$$

(Equivalently, the row sums of the fundamental matrix $N = (I - Q)^{-1} = \begin{pmatrix} 2.083 & 1.25 \\ 0.833 & 2.5 \end{pmatrix}$ are both $\frac{10}{3}$.)

Trap analysis: (A) is $\frac{1}{1-0.4}$: the expected stay at state 2 only, ignoring the trips through state 3. (B) is $N_{22} = 2.083$, the expected visits to state 2 alone. (C) is $N_{33} = 2.5$, the expected visits to 3 from 3: the right number for a different question.

Correct: (D) 3.33

★ *Intuition:* Think of the transient states as rooms in a maze with two exits. The expected escape time is the total time you spend in *every* room along the way — you must add the visits to room 2 and the detours through room 3.

▶ *Speed insight:* Always write "1+" on the right of each first-step equation: it counts the step you are about to take. The fundamental matrix $N = (I - Q)^{-1}$ gives expected visits N_{ij}; its row sums are expected absorption times.

Problem H6 — A reflecting walk where the wall pushes you in

A walk on $\{0, 1, 2, \dots\}$ moves right with probability 0.4, left with probability 0.6 from every state $k \geq 1$, and from 0 moves to 1 with probability *one*. Find the expected return time to 0.

Options: (A) 1.5 (B) 2.5 (C) 3 (D) 6

Solution. Detailed balance across each doorway: $\pi_0 \cdot 1 = \pi_1 \cdot 0.6$, and for $k \geq 1$, $\pi_{k+1} = \pi_k \frac{0.4}{0.6} = \frac{2}{3}\pi_k$. So $\pi_1 = \frac{\pi_0}{0.6}$ and

$$\sum_{k \geq 1} \pi_k = \frac{\pi_0}{0.6} \cdot \frac{1}{1 - \frac{2}{3}} = 5\pi_0 \;\Rightarrow\; 6\pi_0 = 1, \quad \pi_0 = \tfrac{1}{6}.$$

$$\mathbb{E}[\text{return to } 0] = \frac{1}{\pi_0} = \boxed{6}.$$

(Drift is toward 0, so the chain is positive recurrent and π exists.)

Trap analysis: (C) is the answer to the *lazy-wall* version (stay at 0 with probability 0.6): there $\pi_0 = 1 - \rho = \frac{1}{3}$, giving 3. Here the wall has no self-loop, which changes π_0. (A) and (B) are shortcuts such as $\frac{1}{0.4} = 2.5$ that ignore the long excursions into $\{1, 2, \dots\}$.

Correct: (D) 6

★ *Intuition:* A shopper who leaves the front door *always* walks inward, and is nudged back toward the door only 60% of the time. The door has no "loitering" mass (no self-loop), so the shopper spends a smaller share of time at it ($\frac{1}{6}$), and the visits are rarer: one every 6 steps.

▶ *Speed insight:* Mean return time to a state is $1/\pi_i$. Always rebuild the first balance equation from the wall's *actual* transition (stay vs. push-in); a different boundary rule changes π_0 completely.

III. Periodicity

Problem H7 — The last gap in return times (a Frobenius problem)

From state 0 the only cycles are a 4-step loop ($0 \to 1 \to 2 \to 3 \to 0$) and a 5-step loop ($0 \to 4 \to 5 \to 6 \to 7 \to 0$), each taken with probability $\frac{1}{2}$ at 0. What is the *largest* n for which $P^n_{00} = 0$?

Options: (A) 9 (B) 11 (C) 12 (D) 20

Solution. $P^n_{00} > 0$ exactly when $n = 4a + 5b$ for some integers $a, b \geq 0$. Since $\gcd(4, 5) = 1$ the chain is aperiodic, and the largest non-representable n is the Frobenius number

$$4 \cdot 5 - 4 - 5 = \boxed{11}.$$

The zeros of P^n_{00} are $n = 1, 2, 3, 6, 7, 11$; every $n \geq 12$ is reachable ($12 = 4 \cdot 3$, $13 = 4 \cdot 2 + 5$, $14 = 4 + 5 \cdot 2$, $15 = 5 \cdot 3$, and then add 4s).

Trap analysis: (C) is 12, the *smallest* n from which all returns are possible ("$P^n > 0$ for all $n \geq 12$"). (D) is $4 \cdot 5 = 20$, the product, which overshoots. (A) is $4 + 5$, the sum.

Correct: (B) 11

★ *Intuition:* You have 4-cent and 5-cent stamps and want to make exact postage. Every amount from 12 cents up works, but 11 cannot be made. Returning to 0 at time n is the same as making n cents of postage out of 4s and 5s.

▶ *Speed insight:* For two cycle lengths a, b with gcd = 1, the last impossible return time is $ab - a - b$. If $\gcd(a, b) = d > 1$ the chain has period d and returns only at multiples of d.

Problem H8 — A limit along the even times

A walker moves on the path graph $1-2-3-4$, choosing a uniformly random neighbour each step (from the ends the only move is inward). Starting at state 2, find $\lim_{n\to\infty} P(X_{2n}=2)$.
Options: (A) $\frac{1}{3}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) No limit exists
Solution. The graph is bipartite ($\{1,3\}$ vs. $\{2,4\}$), so the chain has period 2. Degrees are $(1,2,2,1)$, so $\pi=\left(\frac{1}{6},\frac{1}{3},\frac{1}{3},\frac{1}{6}\right)$. At even times the walker is confined to $\{2,4\}$, whose stationary mass is $\pi_2+\pi_4=\frac{1}{2}$. Rescaling to a probability on that set:

$$\lim_{n\to\infty} P_{22}^{2n} = 2\pi_2 = \boxed{\frac{2}{3}}.$$

(Numerically $P_{22}^{20}=0.66667$ while $P_{22}^{41}=0$.)
Trap analysis: (A) is π_2, the *long-run fraction of time* at state 2, which is also correct but answers a different question (averaging over odd and even times). (D) says the limit does not exist: P_{22}^n indeed has no limit, but the *subsequence* P_{22}^{2n} does. (B) splits evenly between 2 and 4, ignoring that state 2 has twice the degree of an end.
Correct: (C) $\frac{2}{3}$
★ *Intuition:* A pendulum swings between two sets, left and right. At each moment you only know which side the pendulum is on, so the long-run mass is divided *within that side* — doubled relative to the global average.
▶ *Speed insight:* For a chain of period d and i in a cyclic class, $P_{ii}^{nd}\to d\,\pi_i$. The factor d appears because the chain only "visits" the class every d-th step.

IV. Stationary Distributions in a Market Setting

Problem H9 — How much of the time are you holding inventory?

A market maker's inventory takes values $-1,0,+1$ with

$$P=\begin{pmatrix}0.5 & 0.5 & 0\\ 0.3 & 0.4 & 0.3\\ 0 & 0.5 & 0.5\end{pmatrix}$$

(order: $-1,0,+1$). In the long run, what fraction of periods is inventory nonzero?
Options: (A) 0.273 (B) 0.455 (C) 0.545 (D) 0.600
Solution. The chain is a birth–death chain, so use detailed balance: $\pi_{-1}(0.5)=\pi_0(0.3)\Rightarrow\pi_{-1}=0.6\pi_0$, and by symmetry $\pi_{+1}=0.6\pi_0$. Normalizing, $2.2\pi_0=1$:

$$\pi=\left(\tfrac{3}{11},\tfrac{5}{11},\tfrac{3}{11}\right),\qquad P(\text{nonzero})=\frac{6}{11}\approx\boxed{0.545}.$$

Trap analysis: (D) adds the one-step rates out of flat, $0.3+0.3$, and treats them as stationary probabilities, forgetting that long positions are quickly reduced (0.5 per period). (B) is π_0: it is the fraction of time *flat*, i.e. the complement. (A) is a single side ($\frac{3}{11}$), not both.
Correct: (C) 0.545
★ *Intuition:* Each unit of inventory is pulled back to flat with probability 0.5 per period, but flat only leaves with probability 0.6 in total. Flow in and out of the "away from flat" region must balance, which pins the long-run split at $\frac{5}{11}:\frac{6}{11}$.
▶ *Speed insight:* Birth–death chain ⇒ one balance equation per doorway; always check $\sum\pi=1$ last. The expected holding cost per period is $\sum\pi_i c_i$ — here a cost of \$0.20 per unit gives $0.2\cdot\frac{6}{11}\approx\$0.109$ per period.

Problem H10 — Capstone: return time to Bear and the first-step cross-check

A market has regimes Bull, Flat, Bear with

$$P=\begin{pmatrix}0.7 & 0.2 & 0.1\\ 0.3 & 0.4 & 0.3\\ 0.2 & 0.3 & 0.5\end{pmatrix}.$$

Starting from Bear, what is the expected number of days until the market is next in Bear?
Options: (A) 2.00 (B) 3.54 (C) 3.83 (D) 6.67
Solution. Stationary distribution (columns of $\pi=\pi P$, normalized):

$$\pi=\left(\tfrac{21}{46},\tfrac{13}{46},\tfrac{12}{46}\right).$$

The mean return time to Bear is the reciprocal of its stationary probability:

$$m_{\text{Bear}}=\frac{1}{\pi_{\text{Bear}}}=\frac{46}{12}\approx\boxed{3.83}.$$

Cross-check with first-step analysis. Expected times to *hit* Bear from Bull and Flat solve $k_B=1+0.7k_B+0.2k_F$, $k_F=1+0.3k_B+0.4k_F$, giving $k_B=\frac{20}{3}\approx6.67$, $k_F=5$. Then

$$m_{\text{Bear}}=1+0.2k_B+0.3k_F+0.5(0)=1+1.333+1.5=3.83.\ \checkmark$$

Trap analysis: (D) is the *hitting time* of Bear from Bull (6.67), not the return time from Bear. (B) is $1/\pi_{\text{Flat}}$: the right method on the wrong state. (A) is $\frac{1}{1-0.5}$, the expected stay in Bear: it ignores that after leaving you have to find your way back.
Correct: (C) 3.83
★ *Intuition:* Bear occupies about a quarter of all days ($\frac{12}{46}\approx0.26$), so on average there is one Bear day roughly every 3.8 days — exactly the "every fourth day" rhythm of a quarter-frequency event. The first-step calculation is the same fact seen from inside the chain: wait, leave, wander back.
▶ *Speed insight:* Two independent routes to the same number are the best check on a Stage-1 answer: $m_i=1/\pi_i$ and $m_i=1+\sum_{j\neq i}P_{ij}k_j$ must agree. If they do not, one of your balance equations is wrong.

Calibration note: H1, H3, H4, H5, H9 are realistic 90–150s Stage-1 questions once the pattern (closed form, class-exit share, fundamental matrix, birth–death balance) is recognized. H2, H6, H7, H8, H10 need a Bayes-bridge, a changed boundary rule, a Frobenius number, a periodic-subsequence argument, or a two-route cross-check — Stage-2 grade, but each uses only machinery from Chapter 4.1.